

Kolmogorov-Arnold Sieve Networks for High-Dimensional Estimation and Inference

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Latest draft and full code available at <https://github.com/samiabd8/sieveKAN>

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Proposed Framework

We propose the KASN estimator, a novel sieve extremum estimator that is:

- Based on Kolmogorov-Arnold Networks (**KANs**; featuring *learnable* activation functions).
- Designed for nonparametric estimation in high-dimensional (**H-D**) settings involving time series data.

Proposed Framework

We propose the KASN estimator, a novel sieve extremum estimator that is:

- Based on Kolmogorov-Arnold Networks (**KANs**; featuring *learnable* activation functions).
- Designed for nonparametric estimation in high-dimensional (**H-D**) settings involving time series data.
- Not subjected to the Curse of Dimensionality (**CoD**) through sparsity.
- Successful in delivering a **dimension-free** convergence rate of $o_P(n^{-1/4})$, crucial for semiparametric applications.

Dimension-Free Convergence Rates

For semiparametric models, inference on a finite-dimensional parameter α_0 requires estimating an infinite-dimensional nuisance g_0 .

- Minimax rates for $\|\hat{g}_n - g_0\|$ are $O(n^{-m/(2m+d)})$ (Stone, 1982): depend on both m and d .
- Under $d \asymp n^{\zeta_d}$, these rates are *arbitrarily slow* for any fixed smoothness m .

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- Under $d \asymp n^{\zeta_d}$, these rates are *arbitrarily slow* for any fixed smoothness m .
- The threshold $o_P(n^{-1/4})$ is the **canonical** rate sufficient for:
 - 1 Asymptotic normality of plug-in estimators (Chen & Shen, 1998).
 - 2 $\sqrt{n}$ -consistency of DML causal parameters (Chernozhukov et al., 2018).
- KASN achieves this rate *independently* of d , by exploiting the sparse compositional structure.

Dimension-Free Convergence Rates

Example 1 (Partially Linear Regression Model): Suppose $\{Y_t, D_t, W_t\}_{t=1}^n$ satisfies:

$$Y_t = D_t\alpha_0 + g_0(W_t) + U_t, \quad \mathbb{E}[U_t|D_t, W_t] = 0,$$

where D_t is the (treatment) variable of interest and g_0 is an unknown nonparametric function of controls W_t . Assume $g_0 \in W_2^{m+1}$ admits a sparse compositional form (**K.3(a)**), where μ is the distribution of W_t .

Let $\hat{g}_n \in \mathcal{K}_n$ be the sieve KAN estimator for g_0 obtained using $l(Z_t, \theta) = -(Y_t - D_t\alpha - g(W_t))^2/2$. Then, under **Assumptions K.1-4**, we have:

- (i) $\|\hat{g}_n - g_0\|_{L_2(\mu)} = o_P(n^{-1/4})$, and
- (ii) $\sqrt{n}(\hat{\alpha}_n - \alpha_0) \rightsquigarrow \mathcal{N}(0, \Sigma_\alpha)$

Stochastic Equicontinuity: Condition A (Chen & Shen, 1998)

The key mechanism linking the entropy bound to the convergence rate is **stochastic equicontinuity** of the empirical criterion (Norm: $L_2(\mu)$).

Define the **empirical criterion process**:

$$\nu_n(\theta) \equiv \frac{1}{n} \sum_{t=1}^n \left[l(Z_t, \theta) - l(Z_t, \theta_0) \right] - E \left[l(Z_t, \theta) - l(Z_t, \theta_0) \right]$$

The criterion is stochastically equicontinuous over Θ_n if, for every $\varepsilon > 0$:

$$\lim_{\delta \rightarrow 0} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\substack{\theta \in \Theta_n \\ \|\theta - \theta_0\| \leq \delta}} |\nu_n(\theta)| > \varepsilon \right) = 0$$

Stochastic Equicontinuity: Condition A (Chen & Shen, 1998)

Condition A.3 (Chen & Shen, 1998): Let

$\mathcal{F}_n = \{l(\theta, Y_t) - l(\theta_0, Y_t) : \|\theta - \theta_0\| \leq \delta, \theta \in \Theta_n\}$. There exists $\delta_n \in (0, 1)$ such that:

$$\delta_n = \sup \left\{ \delta > 0 : \delta^{-2} \int_{b\delta^2}^{a\delta} \sqrt{H(w, \mathcal{F}_n)} dw \leq C_2 n^{1/2} \right\}$$

- $H(\epsilon, \mathcal{F}_n)$ is the $L_2(\mu)$ -bracketing entropy of $\mathcal{F}_n$.

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- $H(\epsilon, \mathcal{F}_n)$ is the $L_2(\mu)$ -bracketing entropy of $\mathcal{F}_n$.
- **Theorem 3** verifies this integral condition using our entropy bound (**Corollary 2**), yielding $\delta_n = o(n^{-1/4})$.
- Intuitively: controlled complexity of $\mathcal{K}_n$ (via our novel regularization approach) ensures sup of the EP contracts quickly enough to permit inference.

Asymptotic Normality: Condition B (Chen & Shen, 1998)

Example 2 (Average Partial Derivative Estimation). Suppose $\{Y_t, X_t\}$ satisfies:

$$Y_t = f_0(X_t) + U_t, \quad \mathbb{E}[U_t|X_t] = 0, \text{Var}[U_t|X_t] = \sigma^2(X_t) > 0$$

with $\theta_0 \equiv f_0 \in W_2^{m+1}$ admitting a sparse compositional form. Let $\hat{\theta}_n \in \mathcal{K}_n$ be the KASN extremum estimator obtained using $l(Z_t, \theta)$.

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For a covariate $X_{t,j}$ with $j \in \{1, \dots, d\}$ and under Assumptions **K.1-4** with $k = 4$:

$$(i) \quad \|\hat{\theta}_n - \theta_0\|_{L_2(\mu)} = o_P(n^{-1/4}), \text{ and} \quad (1)$$

$$(ii) \quad \sqrt{n} \left(\frac{1}{n} \sum_{t=1}^n \frac{\partial \hat{\theta}_n(X_t)}{\partial x_j} - \mathbb{E} \left[\frac{\partial \theta_0(X_t)}{\partial x_j} \right] \right) \rightsquigarrow \mathcal{N}(0, \Sigma_{\gamma_j}). \quad (2)$$

where Σ_{γ_j} is long-run variance of the score process $\partial \theta_0(X_t)/\partial x_j$.

Asymptotic Normality: Condition B (Chen & Shen, 1998)

Following Chen and Shen (1998, Theorem 2), normality of plug-in estimators $\Gamma(\hat{\theta}_n)$ requires five conditions (B.1–B.5).

Riesz Representation: For a linear functional $\Gamma : \Theta \rightarrow \mathbb{R}$, there exists a unique $v^* \in \bar{V}$ (the completion of $\text{span}(\Theta - \{\theta_0\})$) such that

$$\Gamma'_{\theta_0}[\theta - \theta_0] = \langle \theta - \theta_0, v^* \rangle, \quad \forall \theta \in \Theta$$

The existence of v^* identifies the direction of asymptotic variability in the sieve space.

Asymptotic Normality: Condition B (Chen & Shen, 1998)

Key conditions that rely on $\|\hat{\theta}_n - \theta_0\| = o_P(n^{-1/4})$:

- **B.4** (Stochastic equicontinuity of score):

$$\sup_{\|\theta - \theta_0\| \leq \delta_n} \mu_n(l'_{\theta_0}[\theta - \theta_0, Z_t]) = O_P(\epsilon_n)$$
- **B.5** (CLT at Riesz representer): $\sqrt{n} \mu_n(l'_{\theta_0}[v^*, Z_t]) \xrightarrow{d} N(0, \sigma_v^2)$

Both conditions are verified for KASN using the entropy bound of **Theorem 2** and CLTs for β -mixing sequences.

Contrasting with MLPs

Although DNNs are powerful approximators, their application in H-D settings suffers from:

- The “black box” nature of the MLP architecture.
- A reliance on i.i.d assumptions in the existing literature.
- Slower convergence rates, subject to the CoD even under regularization.

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- Slower convergence rates, subject to the CoD even under regularization.

CoD: dictates n required for a given estimation accuracy grows **exponentially** with d .

Minimax optimal rates: convergence rates tied to both d , smoothness parameter m (Stone, 1982).

Related Literature

This framework spans several areas of the literature:

- ReLU DNNs: Yarotsky (2017, 2018), Farrell et al. (2021), Schmidt-Hieber (2020, 2021).
- KANs: Liu et al. (2024), Zhang and Zhou (2024).

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- KANs: Liu et al. (2024), Zhang and Zhou (2024).
- Sieve Networks: Hornik et al. (1989), White (1990), Chen and White (1999).
- H-D Estimation: Chernozhukov et al. (2018), Friedman (1991), Gouriéroux and Monfort (2025).

First framework to integrate the method of sieves with modern DNN architectures.

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Optimization Problem

For a stationary process $Z_t \equiv \{Y_t, X_t\}_{t=1}^n$ and given a probability measure P_0 , define:

$$\theta_0 \equiv \mathcal{T}(P_0) \equiv \arg \sup_{\theta \in \Theta} \mathbb{E}_{P_0}[l(Z_t, \theta)] \quad (3)$$

$$\hat{\theta}_n \equiv \arg \sup_{\theta \in \Theta_n} \frac{1}{n} \sum_{t=1}^n l(Z_t, \theta) \equiv \arg \sup_{\theta \in \Theta_n} \mathcal{L}_n(\theta) \quad (4)$$

where:

- $l(\cdot)$ is a quasi-log-likelihood function belonging to a class of general loss functions, e.g., quadratic and quantile.
- $\theta_0 \in \Theta$ (possibly infinite-dimensional space) and $\theta_n \in \Theta_n$ (sieve space dense in Θ : $\Theta_1 \subset \Theta_2 \subset \dots \subset \Theta_n$ and $\|\hat{\theta}_n - \theta_0\| = 0$ as $n \rightarrow \infty$).

Function Spaces and Norms

- Target function assumed to belong in a Sobolev space: $f_0 \in \mathcal{F}_d^{m+1}$
- In nonparametric regression, $\mathcal{F}_d^{m+1} \equiv \Theta$; equipped with the pseudometric $\|\cdot\|$
 - Semiparametric example: $\theta_0 \equiv (\alpha_0, f_0) \in \mathcal{A} \times \mathcal{F}_d^{m+1} \equiv \Theta$

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 - Semiparametric example: $\theta_0 \equiv (\alpha_0, f_0) \in \mathcal{A} \times \mathcal{F}_d^{m+1} \equiv \Theta$
- Establish convergence in the $L_2(\mu)$ norm:

$$\|h\|_{L_2(\mu)} = \left(\int_{\mathbb{R}^d} |h(x)|^2 d\mu(x) \right)^{1/2}$$

weaker than convergence in the Sobolev norm ($L_2(\mu)$): Hilbert space of all measurable functions $h : \mathbb{R}^d \rightarrow \mathbb{R}$ such that $\int_{\mathbb{R}^d} |h(x)|^2 d\mu(x) < \infty$).

- Allows for smoothness parameter m to be absorbed by constants in the resulting rate, sufficient for applications of interest.

KANs: Universal Approximators

Kolmogorov-Arnold Representation Theorem: for a smooth $f : [0, 1]^d \rightarrow \mathbb{R}$,

$$f(X_t) = f(X_{t,1}, X_{t,2}, \dots, X_{t,d}) = \sum_{i=1}^{2d+1} \Phi_i \left(\sum_{j=1}^d \phi_{i,j}(X_{t,j}) \right) \quad (5)$$

- Any continuous multivariate function can be exactly represented by a finite superposition of continuous univariate (“1D”) functions.

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- Any continuous multivariate function can be exactly represented by a finite superposition of continuous univariate (“1D”) functions.
- Non-constructive existence proof, usefulness in the context of networks debated (e.g., Girosi and Poggio, 1989) extensively.
- Liu et al. (2024): KART can be generalized to arbitrary widths and lengths using *matrices* of 1D functions.

KAN Sieve Extremum Estimator

Key assumptions:

- $f_0 \in \mathcal{F}_d^{m+1}$ admits a sparse compositional form: $s_0 \ll 2d + 1$ where s_0 grows slowly.
- The strictly stationary process $\{Z_t\} = \{Y'_t, X'_t\}'$ is:
 - i.i.d or m -dependent (for $m \in \mathbb{N}$), or:
 - β -mixing stationary with $\beta(j) \leq \beta_0 j^{-\xi}$, $\xi > 2$ for some $\beta_0 > 0$, $\xi > 2$.

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 - β -mixing stationary with $\beta(j) \leq \beta_0 j^{-\xi}$, $\xi > 2$ for some $\beta_0 > 0$, $\xi > 2$.
- 1D B-spline functions $(\Phi_i, \phi_{i,j}) \in W_2^{m+1}(\mathcal{I}_\epsilon)$ are Lipschitz continuous with uniformly bounded derivatives. Furthermore, Φ_i admits a composition $\Phi_i = \psi_{i,L_n} \circ \psi_{i,L_n-1} \circ \cdots \circ \psi_{i,1}$.
- Standard regularity conditions on $l(\cdot)$, Θ_n and Θ .

Double/Debiased Machine Learning

Consider the PLR model (Robinson, 1988), given (i.i.d.) $Z_i = \{Y_i, D_i, X_i\}$:

$$Y_i = D_i\alpha_0 + g_0(X_i) + U_i, \quad E[U_i|X_i, D_i] = 0 \quad (6)$$

$$D_i = m_0(X_i) + V_i, \quad E[V_i|X_i] = 0 \quad (7)$$

- Eliminates regularization bias arising from ML estimators of nuisance functions $\eta_0 = (g_0, m_0)$, even when $d \gg n$.

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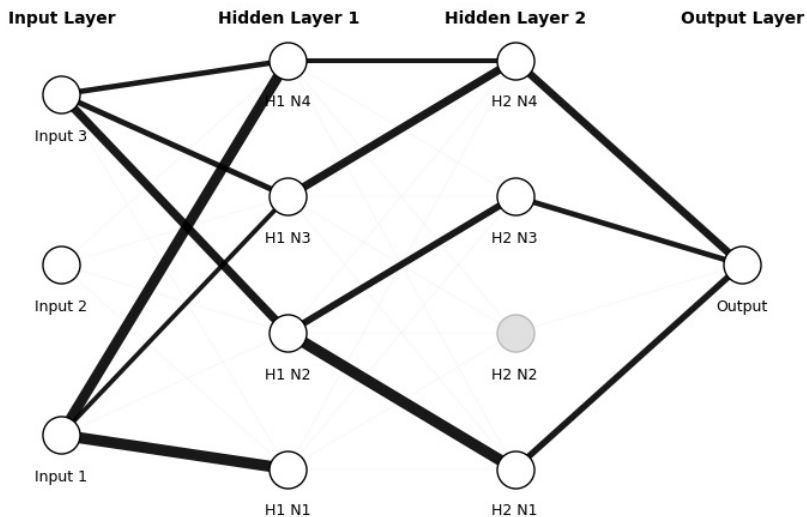
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- Eliminates regularization bias arising from ML estimators of nuisance functions $\eta_0 = (g_0, m_0)$, even when $d \gg n$.
- Obtain $\sqrt{n}$ -consistent and asymptotic normal estimator $\hat{\alpha}$ using Neyman Orthogonality and Sample Splitting.
- Extension of Sample Splitting to stationary mixing processes is left to future work (Abdurahman and Gradojevic, 2026).

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Sieve KAN Active Edges



Sieve KAN Architecture

Key parameters in sieve KAN:

- Network **width** W_n ; corresponding to the number of 1D functions in each hidden layer.
- Number of **layers** L_n ; also define $r_n \equiv L_n W_n^2$.
- Size of the B-spline **grid** G_n .

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$$\text{KAN}(\mathbf{x}) = (\Phi_{L_n-1} \circ \Phi_{L_n-2} \circ \dots \circ \Phi_1 \circ \Phi_0) \mathbf{x} \quad (8)$$

$$\text{MLP}(\mathbf{x}) = (\mathbf{W}_{L-1} \circ \sigma \circ \mathbf{W}_{L-2} \circ \sigma \circ \dots \circ \mathbf{W}_1 \circ \sigma \circ \mathbf{W}_0) \mathbf{x} \quad (9)$$

where σ is a fixed activation function, e.g., ReLU: $\sigma(x) = \max\{0, x\}$.

Use B-spline coefficient constraint (under Group Lasso) to define an adaptive threshold for what constitutes an “active” edge: Δ_n/r_n .

Corresponds to entropy regularization.

Sieve KAN Architecture

Given B-spline order $k \geq m + 1$ and coefficient vector $\mathbf{c}_n \equiv \{\mathbf{c}_j\}_{j=1}^{r_n}$, the finite-dimensional sieve space is given by $\Theta_n \equiv A \times \mathcal{K}_n$, where the KASN class is defined as:

$$\mathcal{K}_n = \left\{ f(\tilde{X}_t; \mathbf{c}_n) \mid f(\mathbf{x}; \mathbf{c}_n) = \Phi_{L_n} \circ (\Phi_{L_n-1} + h) \circ \cdots \circ (\Phi_1 + h) \circ \Phi_0(\mathbf{x}), \right. \\ \left. \mathbf{c}_n \in \mathbb{R}^{r_n \times G_n}, \sum_{j=1}^{r_n} \|\mathbf{c}_j\|_2 \leq \Delta_n \right\}, \quad (10)$$

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Where the identity mapping $h : \mathbb{R}^{W_n^{l-1}} \mapsto \mathbb{R}^{W_n^l}$ is a residual connection $h(x) = x$ between hidden layers. A CDF normalization is applied to inputs $X_t := [X_{t,1}, \dots, X_{t,d}]' \in \mathbb{R}^d$ such that:

$$\tilde{F}_n^j(z) = \frac{1}{n} \sum_{t=1}^n \mathbf{1}\{X_{t,j} \leq z\}, \quad \tilde{X}_{t,j} = \tilde{F}_n^j(X_{t,j}), \quad j = 1, \dots, d \quad (11)$$

Sieve KAN Architecture

We define KASN parameters $p_n \equiv (L_n, W_n, G_n)$ as follows:

$$L_n \asymp \log n, \quad W_n \gtrsim s_0 \asymp \log n, \quad G_n \asymp n^\gamma, \quad \Delta_n \asymp n^{\zeta_\delta}$$

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Key conditions ($\zeta_\delta > 0$):

$$\bar{s}_0 \asymp s_0 \times L_n = O((\log n)^2), \quad \frac{1}{4k} < \gamma < \frac{1}{2} \quad (k \geq m+1) \quad (12)$$

The upper bound $\gamma < \frac{1}{2}$ arises directly from the entropy integral in Condition A.3 of Chen & Shen (1998), lower bound from **Theorem 1**.

By letting $p_n \rightarrow \infty$ and $\Delta_n \rightarrow \infty$ as $n \rightarrow \infty$, we obtain a *sequence of increasingly flexible* network architectures; analogous to White (1990).

Compactness of Sieve Space

Constrained estimator: $\hat{\theta}_n = \arg \sup_{\theta \in \mathcal{K}_n} \mathcal{L}_n(\theta)$, where the sieve space imposes:

$$\theta \in \mathcal{K}_n \iff \sum_{j=1}^{r_n} \|\mathbf{c}_j\|_2 \equiv \|\mathbf{c}_n\|_{1,2} \leq \Delta_n, \quad \Delta_n \asymp n^{\zeta_\delta}$$

Compactness: $\mathcal{K}_n$ is deterministic (given Δ_n) and compact: the $\ell_{1,2}$ ball is closed and bounded, and continuity of $\mathcal{L}_n(\theta)$ via **K.2(c)** guarantees exact attainment of the supremum by the *extreme value theorem*.

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Penalized equivalence: In practice, optimize the penalized objective $\mathcal{L}_n(\theta) - \lambda \|\mathbf{c}_n\|_{1,2}$. By **Lagrangian duality** (under **K.2(b)-(c)**), the two problems are equivalent at

$$\|\hat{\mathbf{c}}(\lambda_n)\|_{1,2} = \Delta_n, \quad \lambda_n \equiv \inf \{ \lambda > 0 : \|\hat{\mathbf{c}}(\lambda)\|_{1,2} \leq \Delta_n \}$$

Cross-validation: λ_n is selected via CV over the monotone correspondence $\lambda \mapsto \|\hat{\mathbf{c}}(\lambda)\|_{1,2}$, which is equivalent to selecting ζ_δ directly.

Critique: Gouriéroux and Monfort (2025)

Two main critiques raised in Gouriéroux and Monfort (2025):

- (i) KART is non-unique at the component level (w.r.t. $\phi_{i,j}$ and Φ_i)
- (ii) CoD poses a threat to identification with MLPs and KANs

(i): Addressed by identification of θ_0 under QMLE. Compactness of $\mathcal{K}_n$ and **K.2(b)** guarantee the existence of a measurable maximizer $\hat{\theta}_n$.

(ii): **Corollary 3** establishes that the convergence rate $o_P(n^{-1/4})$ holds uniformly over $d = O(n^{\zeta_d})$, rendering dimensional scaling irrelevant to the estimation problem.

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Theoretical Results: Approximation Error

Theorem 1 (Approximation Error): Under Assumptions **K.1–K.4**, there exists a sieve approximation $\pi_n \theta_0 \in \mathcal{K}_n$ such that:

$$\|\pi_n \theta_0 - \theta_0\| \leq C_{\bar{s}_0} G_n^{-k} + C'_{\bar{s}_0} O(n^{-1/2}) = o(n^{-1/4}),$$

where $C_{\bar{s}_0} > 0$ depends on $\bar{s}_0 = s_0 \times L_n$, Lipschitz constants (B, L) , and spline order $k \geq m + 1$.

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where $C_{\bar{s}_0} > 0$ depends on $\bar{s}_0 = s_0 \times L_n$, Lipschitz constants (B, L) , and spline order $k \geq m + 1$.

Corollary 1 (Sobolev Norm Bound): Under the same conditions with $k = 4$:

$$\|\pi_n \theta_0 - \theta_0\|_{W_2^{m+1}(\mu)} \leq C_{\bar{s}_0} G_n^{-1} + C'_{\bar{s}_0} O(n^{-1/2}) = o(n^{-1/4}),$$

with $\gamma > \frac{1}{4}$ sufficient for all $m \leq 3$. Relevant for **average derivative** plug-in functionals (Example 2).

Theoretical Results: Entropy Bounds

Theorem 2 (Entropy Bound): Under Assumptions **K.1–K.4**, given sieve space $\mathcal{K}_n$ with $s_n \geq \bar{s}_0$:

$$\log \mathcal{N}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L_2(\mu)}) \leq C_1 s_n \log\left(\frac{r_n}{s_n}\right) + C_2 s_n G_n \log\left(\frac{\Delta_n}{\epsilon}\right) + o(1)$$

where the $o(1)$ term accounts for the (fixed) residual connection mapping h , which contributes a unit covering number.

Corollary 2 (Bracketing Entropy): Under Assumptions **K.1–K.4**:

$$\log \mathcal{N}_{[]}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L_2(\mu)}) \leq C_1 s_n \log\left(\frac{r_n}{s_n}\right) + C_2 s_n G_n \log\left(\frac{2\Delta_n}{\epsilon}\right) + o(1)$$

Used directly to verify Condition A.3 of Chen and Shen (1998) in the proof of **Theorem 3**.

Theoretical Results: Estimation Error

Theorem 3 (Estimation Error): Under Assumptions **K.1–K.4**, given sieve space $\mathcal{K}_n$ and letting $\frac{1}{4k} < \gamma < \frac{1}{2}$:

$$\|\hat{\theta}_n - \pi_n \theta_0\| = o_P(n^{-1/4})$$

Corollary 3 (Convergence Rate): Under Assumptions **K.1–K.4**, the KASN sieve extremum estimator achieves:

$$\|\hat{\theta}_n - \theta_0\| = o_P(n^{-1/4})$$

independently of $d = O(n^{\zeta_d})$ for any $\zeta_d > 0$.

Theoretical Results: Asymptotic Normality

Theorem 4 (Asymptotic Normality): Under Assumptions **K.1–K.4** and **B.1–B.5** (Chen & Shen, 1998), the plug-in KASN estimator satisfies:

$$\sqrt{n}\left(\Gamma(\hat{\theta}_n) - \Gamma(\theta_0)\right) \rightsquigarrow N(0, \sigma_*^2) \quad (13)$$

where $\sigma_*^2 = \text{Var}(l'_{\theta_0}[v^*, Z_1]) + 2 \sum_{j=2}^{\infty} \text{Cov}(l'_{\theta_0}[v^*, Z_1], l'_{\theta_0}[v^*, Z_t])$

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- $v^* \in \bar{V}$: the Riesz representer of the linear functional Γ .
- Long-run variance $\sigma_*^2 > 0$ guaranteed by $v^* \neq 0$ under **K.2(a)**.
- B.5 holds via CLT for β -mixing sequences (Yokoyama, 1980).
- In ultra-high-dimensional regimes ($d \gg n$), regularization bias may affect $\Gamma(\hat{\theta}_n)$ directly (use DML!).

DML Application

Definition 1 (Neyman Orthogonality): The score $\psi(\cdot; \cdot)$ satisfies Neyman orthogonality if:

$$\partial_{\eta} E[\psi(Z; \alpha_0, \eta_0)][\eta - \eta_0] = 0 \quad \text{for all } \eta \in \mathcal{T},$$

where $\mathcal{T} \equiv \Theta$ for simplicity.

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Proposition 1 (DML Estimation): Under Assumptions **K.1(a)**-**K.4**, the DML estimator $\hat{\alpha}$ given $(\hat{g}_n, \hat{m}_n) \in \mathcal{K}_n$ satisfies:

$$\sigma^{-1} \sqrt{n}(\hat{\alpha} - \alpha_0) \rightsquigarrow N(0, 1), \quad \sigma^2 = (E[V^2])^{-1} E[V^2 U^2] (E[V^2])^{-1}$$

which holds when σ^2 is replaced with $\hat{\sigma}^2$ (Theorem 4.1, Chernozhukov et al., 2018).

Monte Carlo Simulations

We begin with a simple nonparametric regression model:

$$f_0(X) = \sum_{j=1}^S \sin(X_j) + \epsilon, \quad \epsilon \sim N(0, 0.25) \quad (14)$$

With $n = 5000$ observations and $S = 30$ out of $d = 100$ total covariates that are relevant to the target function.

Performance of the sieve KAN is compared against the single (hidden) layer feedforward network (SLFN) from Chen and White.

Monte Carlo Simulations

We set $W_n = 201$ to highlight the sieve KAN performance under significant overparameterization (with $\gamma = 0.499$, $\zeta_\delta = 0.9$) and run $N = 1000$ simulations:

Table: Nonparametric Regression Results

Metric (Avg)	sKAN	SLFN
Train RMSE	1.66	2.34
Train R^2	0.83	0.66
Test RMSE	1.87	2.39
Test R^2	0.78	0.64
Runtime (s)	2513.88	0.33
Sparsity (%)	97.61	N/A

Monte Carlo Simulations

For our DGP simulations, we have: $\alpha_0 = 1.5$, $n = 1000$, $L_n = 6$, $d = 100$ and $S = 30$. The (compositional) nuisance functions are defined as:

$$m_0(\mathbf{X}) = \sin\left(\sum_{j=1}^S \frac{1}{1 + e^{-X_j}}\right) + 0.1 \sum_{j=1}^S X_j^2 + \epsilon, \quad \epsilon \sim N(0, 0.25) \quad (15)$$

$$g_0(\mathbf{X}) = \frac{1}{1 + e^{-\sum_{j=1}^S \sin(X_j)}} + 0.1 \sum_{j=1}^S |X_j| + \nu, \quad \nu \sim N(0, 0.25) \quad (16)$$

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Where residuals $\hat{V} = D - \hat{m}(\mathbf{X})$, and $\hat{U} = Y - \hat{g}(\mathbf{X})$ are used to obtain

$$\hat{\alpha} = \frac{\sum \hat{V}_i (Y_i - \hat{g})}{\sum \hat{V}_i D_i}$$

Monte Carlo Simulations

Here, $W_n = 201$ for sieve KAN #1 ($\gamma = 0.5$, $\zeta_\delta = 0.9$) and $W_n = 101$ for sieve KAN #2 ($\gamma = 0.35$, $\zeta_\delta = 0.75$), $N = 1000$ simulations with 5 folds:

Table: DML Simulation Results: ($\alpha_0 = 1.5$)

Metric (Avg)	sKAN #1	sKAN #2	SLFN	Lasso
Causal Parameter ($\hat{\alpha}$)				
$\hat{\alpha}$	1.502	1.480	1.666	1.732
RMSE	0.066	0.046	0.167	0.232
Runtime (s)	2302.98	1070.20	0.17	0.04
Nuisance Function Performance				
Train R^2 ($\hat{m}_0$ / $\hat{g}_0$)	0.80 / 0.82	0.84 / 0.85	0.09 / 0.07	0.18 / 0.16
Test R^2 ($\hat{m}_0$ / $\hat{g}_0$)	0.39 / 0.45	0.36 / 0.43	-0.04 / -0.06	-0.13 / -0.17
Sparsity % ($\hat{m}_0$ / $\hat{g}_0$)	0.925 / 0.929	0.888 / 0.890	NA / NA	NA / NA

Empirical Application #1

Consider the Sparse Macro-Finance Factors dataset from Rapach and Zhou (2025):

- $d = 118$ macroeconomic and financial variables from the FRED-MD monthly database (1963-2024)
- Small sample size ($n = 726$) and large covariate dimension provides an ideal finite-sample performance test
 - Rapach and Zhou (2025) use sparse PCA to obtain 18 key components (e.g., housing, yields, credit spreads)
 - Findings suggest generally linear underlying relationships.
- Apply a subset of key estimated linear components with $W_n = 243$.

Empirical Application #1

Table: Sparse Macro-Finance Factors

Metric	LASSO	Sieve KAN
Train RMSE	4.49	3.94
Train R^2	0.38	0.52
Test RMSE	4.70	4.75
Test R^2	0.52	0.51
Training Time (s)	1.85	534.46
Sparsity (%)	89.3	97.7

Empirical Application #2

We evaluate KASN on forecasting the 1-year constant maturity Treasury yield (GS1) from FRED-MD, using a 12-month lag structure:

Table: Forecasting Exercise: (12-Month Lag, GS1)

Metric	LASSO	SLFN	KASN
Train RMSE	0.4525	0.3559	0.0265
Test R^2	0.1154	-0.2200	0.5266
Sparsity (%)	88.89	N/A	98.32
Training Time (s)	1.40	9.76	117.03
DM Statistic (vs. KASN)	2.0695	2.5543	N/A
p-value (5% sig.)	0.0385	0.0106	N/A

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Future Work

Extending to activation functions belonging to a RKHS:

- Mild structural assumptions on function space, permits even weaker assumptions on $\phi(\cdot)$.
- Potentially more computationally robust method, encompasses current approach.
- Generalizes beyond B-splines (ex. wavelets) naturally.

Future Work

Most recent draft and full code available at:
<https://github.com/samiabd8/sieveKAN>



Sieve KAN CDE Example

Consider the following function:

$$f_0(x_1, x_2, \dots, x_{10}) = -3.0 + 2 \sin(x_1) + 0.1(x_2^2) + \epsilon, \quad \epsilon \sim N(0, 0.25)$$

where true density takes on a bullseye shape, $d = 10$ and $n = 10,000$.

